

SHAYVERI CORE - Pre-Phase 1 Summary

Compute Orchestration & Rendering Engine · Java / Spring Boot backend

Everything done and everything to understand before Module 1 is written. July 2026.

What This Document Is

A complete recap of the project up to the start of Phase 1: what was scaffolded, why it is shaped the way it is, what every folder is for, what git ignores and why, and the current git state. The build plan (the seven modules, milestones, acceptance criteria) is deliberately NOT in this document. It lives independently in README.md at the project root and as Idea_Generation/docs/plan.txt, and stands as its own plan.

Project location:

```
/Users/averi/Undergrad/CS/Java/Java projects/  
SHAYVERI CORE (Compute Orchestration & Rendering Engine)
```

1. What Was Built (Phase 0 Only)

Phase 0 is deliberately just the skeleton: the app boots, API-key security works, and one ping endpoint exists. No telemetry, jobs, nodes, or config logic has been written yet. That is Phases 1 through 4.

Files created

File	Purpose
build.gradle.kts / settings.gradle.kts	Gradle Kotlin DSL build. Spring Boot 3.5.0, Java 21 toolchain, starters: web, websocket, data-mongodb, data-redis, validation, actuator, security.
gradlew, gradlew.bat, gradle/wrapper/	Committed Gradle wrapper pinned to 8.14.2 so anyone can build without installing Gradle.
docker-compose.yml	Local dev infra: mongo:7 and redis:7-alpine, one command (docker compose up -d).
src/main/resources/application.yml	Mongo/Redis URIs, virtual threads enabled, three API keys from env vars with dev defaults, Open Cloud key/universe placeholders, actuator exposure.
CoreApplication.java	Entry point. Three lines: main() calls SpringApplication.run().
common/PingController.java	GET /api/ping returns {"service":"shayveri-core","status":"ok"}.
common/ApiKeyAuthFilter.java	OncePerRequestFilter reading the X-API-Key header and mapping it to a role.
common/ApiKeyProperties.java, ApiKeyRole.java	Loads the three keys from config; ROBLOX / NODE / DASH role enum.
config/SecurityConfig.java	Stateless, CSRF off, /actuator/health public, everything else requires a valid key.
README.md	The full build plan (system context, all 7 module specs, milestones) as project docs.
.gitignore	Excludes .gradle/, build/, .idea/, *.iml and other generated files. Section 5 covers every entry in depth.

2. Decisions and Deviations From the Plan

Spring Boot 3.5.0 instead of the pinned 3.3.4

The plan pinned Boot 3.3.4, but that version has dropped out of Spring Initializr's supported range (it now requires 3.5.0 or newer). Boot 3.5.x satisfies the same 'Boot 3.3+' requirement.

Hand-built scaffold instead of the start.spring.io curl command

The hosted Spring Initializr service was returning HTTP 500 errors server-side, even for minimal requests, so the project was scaffolded manually with identical results (same dependencies, layout, and config).

Gradle installed via Homebrew (9.6.1)

Installed only to generate the wrapper; the project itself uses the committed wrapper (8.14.2), so the global install is not a build dependency.

Storage behind interfaces (added after the asdb discussion)

Each module will define a small store interface for its persistence needs (JobStore, HeartbeatStore, ConfigStore); Mongo/Redis implementations are adapters behind it. Business logic never touches a Mongo/Redis API directly, which keeps each storage slot independently swappable, e.g. for asdb (custom Rust database) as a v2 milestone.

3. Security Model

Role	Who	Env var	Dev default
ROBLOX	Roblox game servers	SHAYVERI_KEY_ROBLOX	dev-roblox-key
NODE	Lab node agents	SHAYVERI_KEY_NODE	dev-node-key
DASH	React dashboard	SHAYVERI_KEY_DASH	dev-dash-key

Keys are sent in the X-API-Key header. No sessions, no CSRF, fully stateless. /actuator/health is the only public endpoint.

4. Folder Structure

4.1 The full tree

```
SHAYVERI CORE (...)/
|-- README.md          <- project root
|-- build.gradle.kts    <- the build plan; GitHub renders it on the repo front page
|-- settings.gradle.kts <- WHAT to build: dependencies, Java version, plugins
|-- docker-compose.yml  <- project name (one line)
|-- gradlew / gradlew.bat <- recipe to start local MongoDB + Redis in one command
|-- gradle/wrapper/     <- scripts that download and run Gradle (Mac/Linux + Windows)
|-- .gitignore          <- config + jar the gradlew scripts use; commit, never edit
|-- Idea_Generation/    <- (hidden) tells git which folders to never commit
|   |-- docs/           <- storage: not part of the program
|   |-- auxiliary/      <- the plan, standing alone: plan.txt
|   |-- src/            <- this PDF and future working documents
|   |-- main/java/dev/shayveri/core/ <- ALL the actual code lives under here
|       |-- CoreApplication.java <- production code
|       |-- common/             <- entry point, main()
|       |-- config/             <- shared: API-key auth, ping endpoint
|       |-- ingress/            <- Spring configuration (security rules)
|                               <- Module 1: telemetry intake (empty yet)
```

```

| | | -- nodes/          <- Module 2: lab node registry      (empty yet)
| | | -- jobs/           <- Module 3: job queue + lifecycle  (empty yet)
| | | -- overrides/      <- Module 4: game config overrides  (empty yet)
| | | -- egress/         <- Module 5: Roblox Open Cloud    (empty yet)
| | | -- realtime/       <- Module 6: WebSocket hub      (empty yet)
| | | -- observability/  <- Module 7: metrics + audit log   (empty yet)
| | -- main/resources/   <- non-code files the app reads: application.yml
| | -- test/java/        <- tests, mirroring the main layout (empty so far)
|-- .gradle/             <- (hidden, GENERATED) Gradle's cache: never commit
|-- .idea/               <- (hidden, GENERATED) IntelliJ settings: never commit
|-- build/               <- (GENERATED) compiled output: never commit

```

4.2 Subfolders outside src/ - what each one is for in the program

gradle/

The build system's bootstrap. Its one subfolder, `wrapper/`, holds a tiny jar and a properties file that pin the exact Gradle version (8.14.2). When anyone runs `./gradlew`, the script uses these to download that exact Gradle automatically. Abstracted purpose: it is what makes the build reproducible on any machine with zero setup. Committed once, never edited by hand.

Idea_Generation/

The storage folder; not part of the program, so no abstracted program purpose to explain. Its two subfolders: `docs/` holds `plan.txt`, the plain-text copy of the README build plan (system context, all seven module specs, milestones, acceptance criteria) - the contract every other component (dashboard, node agents, Roblox scripts) is built against; when the plan changes, `README.md` is edited and `plan.txt` is re-synced from it. `auxiliary/` holds this PDF and future working documents and notes.

.gradle/, .idea/, build/ (the generated three)

Machine output, not source. `.gradle/` is Gradle's private working cache for this project; `.idea/` is IntelliJ's per-machine settings (window layout, local run configs); `build/` is where compiled `.class` files and packaged jars land when the project builds. All three are recreated automatically from scratch whenever needed, which is exactly why git ignores them (section 5). Never open, never edit, never commit.

4.3 The dividing line: src/ versus everything else

Everything outside `src/` exists to build, run, document, or support the program: the build definition (the `.kts` files), the build bootstrap (`gradlew` + `gradle/`), the local infrastructure recipe (`docker-compose.yml`), the documentation (`README.md`, `Idea_Generation/`), and machine output (the generated three). None of it ships logic.

Everything inside `src/` IS the program: the Java that compiles into the running service, plus the config files it reads at startup. If the repo were a restaurant, `src/` is the kitchen; everything else is the lease, the supplier contracts, and the instruction manuals.

4.4 Inside src/

The three fixed branches

`src/main/java` holds production code, `src/main/resources` holds non-code files the app reads at startup (`application.yml`), and `src/test/java` holds tests, mirroring the main layout. This split is the standard Gradle/Maven layout: the build tool finds code by location, not by configuration, which is why compilation worked with zero 'where is my code' setup.

Why dev/shayveri/core - packages are folders

Java package names map one-to-one onto folder paths, and the compiler enforces the match: a class declared 'package dev.shayveri.core.common;' must physically live in dev/shayveri/core/common/. The reverse-domain prefix (dev.shayveri, like com.google or org.springframework) guarantees these classes can never collide with a library class that happens to share a name. The middle folders (dev/, shayveri/) never contain files directly; they only spell out the namespace.

The module packages

Under dev.shayveri.core the code is organized package-by-module, not by layer: config/ and common/ hold Spring setup and shared pieces (the API-key filter, the ping endpoint), and the seven module folders (ingress, nodes, jobs, overrides, egress, realtime, observability) each own one module from the plan - its controllers, services, store interfaces, and documents together in one place. All seven are empty until their phase is built. Git does not track empty directories, so they will not appear in the repo until their first class lands.

Making IntelliJ display it sanely

IntelliJ collapses the empty middle folders: the Project panel shows one node 'dev.shayveri.core' instead of three nested folders. If it does not, click the gear icon in the Project panel and enable 'Compact Middle Packages'. Finder is the wrong viewer for a Java project; the IDE view is the real one.

4.5 The mental model - three zones

Zone 1, files you write: everything in src/, build.gradle.kts, settings.gradle.kts, docker-compose.yml, README.md, Idea_Generation/.

Zone 2, files written once and then left alone: gradlew, gradlew.bat, gradle/wrapper/, .gitignore.

Zone 3, machine-generated, never opened, never committed: .gradle/, build/, .idea/.

4.6 Why the root files cannot move into a subfolder

Gradle looks for build.gradle.kts, settings.gradle.kts, gradlew, and gradle/ at the project root, period; git reads .gitignore from the repo root; 'docker compose up' looks for docker-compose.yml in the current directory. These locations are contracts with the tools, not stylistic choices. Every Spring Boot repo on GitHub has this identical root set, and people reading the repo expect it there.

5. The .gitignore, Explained

The .gitignore is a hidden text file at the repo root listing path patterns git must pretend do not exist: they never show in git status, never get staged by git add. The governing principle: commit what humans write, ignore what machines derive. Anything a tool can regenerate from the committed sources has no business in history - it bloats every clone, changes on every build, and causes merge conflicts that mean nothing. Entry by entry:

.gradle/

Gradle's private per-project cache: dependency metadata, task state, file hashes it uses to decide what needs recompiling. Regenerated on every build, different on every machine.

build/

All build output: compiled .class files, processed resources, test reports, packaged jars. Fully derivable from `src/` by running `./gradlew build`. Committing compiled output alongside source is redundant and drifts out of sync. (This is exactly the junk the drag-and-drop GitHub upload accidentally included.)

`!gradle/wrapper/gradle-wrapper.jar`

The one deliberate exception, marked by the leading `!` which means do-NOT-ignore. Jars are normally build junk, but the wrapper jar is the bootstrap that lets `./gradlew` work on a machine with no Gradle installed, so it must be committed. The `!**/src/main/**/build/` style entries are the same idea: if a source folder legitimately contained a directory named 'build', it would not be swept up by the `build/` rule.

`.idea/`, `*.iml`, `*.iws`, `*.ipr`, `out/`

`.idea/` holds IntelliJ's per-machine project settings: window layout, local run configurations, per-user editor state. The `*.iml/*.iws/*.ipr` patterns are IntelliJ's older module/workspace/project file formats, and `out/` is IntelliJ's own compile output folder (its equivalent of `build/` when you use the IDE's builder instead of Gradle). All of it is personal to one machine and one user; committing it causes pointless conflicts between collaborators' IDE preferences.

`.apt_generated`, `.classpath`, `.factorypath`, `.project`, `.settings`, `.springBeans`, `.sts4-cache`, `bin/`

The same story for Eclipse and Spring Tool Suite users: these are Eclipse's project metadata and output folders. Nobody on this project uses Eclipse today, but the standard Spring Boot gitignore covers it so a future collaborator's IDE files never pollute the repo.

`.vscode/`

VS Code's per-machine workspace settings. Same reasoning as `.idea/`.

`HELP.md`

Spring Initializr drops a boilerplate `HELP.md` with links into generated projects; it is noise, so the standard gitignore excludes it.

`.DS_Store`

macOS Finder's hidden per-folder metadata (icon positions, view options). Appears in any folder you browse; means nothing to anyone else.

6. Why the Code Looks 'Empty'

Spring Boot inverts program flow compared to a plain `Main.java` project: you write small annotated classes (`@RestController`, `@Configuration`) and the framework finds them, wires them together, and calls them when HTTP requests arrive. The server loop, socket handling, and JSON parsing all come from the framework dependencies declared in `build.gradle.kts`. This is the standard style for every Spring Boot backend and was locked in by the plan's stack decisions. Substantial readable logic arrives with Module 1.

The deep folders (`.gradle/`, `build/`, `gradle/wrapper` contents) are build tooling and caches, not source code. They are gitignored and never edited by hand.

7. Verification Status

Verified: `./gradlew compileJava` passes cleanly.

Not yet verified: Docker is not installed on this machine, so docker compose up, ./gradlew bootRun, GET /actuator/health, and the key-gated /api/ping check have not been run against live Mongo/Redis. That is the remaining Phase 0 acceptance work.

8. Git - Current State and What Gets Committed

Where things stand right now

The local repo is initialized with identity configured, and the clean Phase 0 commit exists locally (18 files, branch main, remote wired to github.com/AveriWylie/SHAYVERI-CORE-). The GitHub side currently holds a drag-and-drop web upload instead: everything nested one folder too deep, compiled build/ junk included, hidden files (.gitignore) missing, and gradlew's executable permission lost. The fix is a one-time login then a force push, which replaces the uploaded mess with the clean commit:

```
gh auth login          # GitHub.com > HTTPS > Yes > Login with a web browser
git push --force -u origin main
```

After that, the day-to-day loop is: git add . then git commit -m "message" then git push.

What belongs in a commit

Commit: src/, gradle/wrapper/ (including the jar), gradlew, gradlew.bat, build.gradle.kts, settings.gradle.kts, docker-compose.yml, README.md, Idea_Generation/, .gitignore.

Never commit (and .gitignore enforces it): .gradle/, build/, .idea/, *.iml. If git status ever shows these as addable, something is wrong with the .gitignore.

9. Next Step: Phase 1

Module 1 (Ingress - telemetry intake) plus a minimal Module 6 (Realtime WebSocket hub). Definition of done: a fake Luau script (curl loop) posts telemetry and it is visible on a raw WebSocket client. Full endpoint specs are in the plan: README.md at the root, or Idea_Generation/docs/plan.txt.